QP CODE: **Reg. No. : .....**
Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MG4DSEPHY203 – Optoelectronics

Duration: 1.5 hrs Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions
Answer any **10** questions
Each question carries **2** Mark

1.	Write down the equation governing the probability of finding an electron in a quantum state with energy E.	[K][1]
2.	What is meant by compensation doping?	[U][1]
3.	Distinguish between nondegenerate and degenerate semiconductors.	[A][1]
4.	Define the minority carrier recombination time.	[U][1]
5.	What happens to the built in potential barrier when a reverse biased is applied across a pn junction diode?	[U][1]
6.	What is meant by a heterojunction LED.	[U][2]
7.	Demonstrate how a quantum well LED structure is formed.	[A][2]
8.	Define the internal quantum efficiency of an LED.	[A][2]
9.	Explain the principle of a white light emitting LED.	[U][3]
10.	What is the basic principle of a pn junction photodiode?	[U][3]
11.	What is meant by photovoltaic mode of a photodiode?	[A][3]
12.	Write down the equation governing the V-I characteristic of a photodiode in the dark.	[K][3]
13.	Define the upper cutoff wavelength of a photodetector.	[U][3]
14.	What is meant by shot noise of a photodetector?	[A][3]

[2×10 = 20]

Part B

Short Essay Type Questions
Answer any **6** questions
Each question carries **5** Mark

15. With the help of an energy band diagram, explain the difference between direct and indirect bandgap semiconductors. [A][1]

16. Discuss briefly about the depletion layer capacitance of a pn junction. [U][1]

17. Draw the energy band diagrams for a *pn* junction under (a) open circuit, (b) forward bias, (c) reverse bias conditions [A][1]

18. Briefly explain any two methods for coupling the light from a surface emitting LED to an optical fibre. [A][2]

19. Briefly discuss about LED materials. [U][2]

20. With the help of a schematic, discuss briefly about the working of a pin photodiode. [U][3]

21. With the help of a schematic, discuss briefly about the working of a phototransistor. [U][3]

22. A solar cell having surface area of $0.01 m^2$ is exposed to sunlight of intensity 900 W/m^2. The maximum power output obtained is 16 mW. Short circuit current is 35 mA and open circuit voltage is 0.62 V. Calculate the Efficiency and Fill-factor of the solar cell. [A][3]

[5×6 = 30]